

MD ASHIKUR RAHMAN

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Research Interests

Trustworthy multimodal AI, with emphasis on vision-language grounding and hallucination evaluation, uncertainty-aware language-model systems, and risk-controlled tool use.

Education

American International University-Bangladesh

2011-2015

B.Sc. in Computer Science and Engineering

- CGPA: 3.87/4.00; top 3%; magna cum laude
- Merit Scholarship and Tuition Fee Waiver

Peer-Reviewed Publications

[P1] Abdullah Ibne Hanif Arean, Niamul Hassan Samin, Md Arifur Rahman, Renu Akter Sweetty, Juena Ahmed Noshin, **Md Ashikur Rahman***. *Stroke-Level Connectivity Verification: Grounding Vision-Language Models Against Topology Hallucination in Diagram Understanding*. *International Conference on Document Analysis and Recognition (ICDAR)*, 2026. Accepted for oral presentation.

*Corresponding author. [Code/Dataset](#)

Preprints and Manuscripts Under Review

[M1] **Md Ashikur Rahman**, Md Arifur Rahman, Niamul Hassan Samin, Khandaker Rifah Tasnia, Sifat Rahman Ahona, Juena Ahmed Noshin. *Beyond Aggregate Risk: Role-Stratified Conformal Risk Control for LLM Tool Calls*. Manuscript under review, 2026.

[M2] **Md Ashikur Rahman**, Juena Ahmed Noshin, Niamul Hassan Samin, Abdullah Ibne Hanif Arean, Md Hasibul Amin, Md Arifur Rahman. *When Detector-Based Grounding Metrics Measure Vocabulary: A Cautionary Audit of Entity Claims in Video-QA Reasoning Traces*. Manuscript under review, 2026.

[M3] **Md Ashikur Rahman**, Md Arifur Rahman, Nusrat Jahan Trisna, Juena Ahmed Noshin. *Math-Encoded Jailbreaks Across Provider-Matched Models and Inference-Time Reasoning Configurations*. Manuscript under review, 2026.

Additional Publication

[A1] *Automated Detection of Diabetic Retinopathy Using Deep Residual Learning*. *International Journal of Computer Applications*, 2020.

Research Experience

Researcher, Trustworthy Multimodal AI - The KOW Company

Jun 2025-Present

- Designed a role-stratified conformal risk-control framework for heterogeneous LLM tool calls, targeting reliability beyond aggregate risk metrics [M1].
- Conducted a cautionary audit of detector-based grounding metrics on Video-QA reasoning traces, showing that such metrics can track vocabulary rather than visual grounding [M2].
- Designed evaluations of math-encoded jailbreaks across provider-matched models and inference-time reasoning configurations [M3].
- Developed stroke-level connectivity verification for detecting topology hallucinations in diagram understanding; served as last and corresponding author on an ICDAR 2026 oral paper [P1].
- Built evaluation frameworks, benchmarks, and reproducible pipelines for multimodal grounding, hallucination analysis, and model reliability.

Research Mentoring

- Mentor and supervise four researchers at The KOW Company on literature review, research planning, experimental design, reproducible implementation, result analysis, and academic writing.
- Advise junior collaborators through manuscript preparation and submission for projects spanning VLM grounding, LLM safety, and risk-controlled tool use.

Open-Source Research Artifacts

- **LogicBench-1K** - Diagram benchmark and supporting artifacts for evaluating topology hallucination and structural grounding; released with the ICDAR 2026 paper [P1]. Role: research lead / corresponding author; contributions include problem formulation, benchmark design oversight, evaluation methodology, and manuscript leadership.

Industry Experience

Lead AI Engineer, The KOW Company

Jan 2023-Present

- Lead development and deployment of multimodal AI systems spanning image/video generation, virtual try-on, computer vision, and 3D reconstruction (Retouched.ai, Omnimage.ai, HoloSnap.ai, CogniX, The Fitting Room).
- Provide technical leadership across machine-learning architecture, offline evaluation, production deployment, and cross-functional delivery.

Senior Machine Learning Engineer, The KOW Company

Jul 2021-Dec 2022

- Improved object-detection and segmentation performance by 20-35% on internal benchmarks; models deployed via Retouched.ai.
- Led 6+ client ML engagements and built offline evaluation and production A/B testing workflows for model quality and reliability.

Machine Learning Engineer, The KOW Company

Jul 2020-Jun 2021

- Developed production deep-learning models and scalable training/evaluation pipelines for object recognition, segmentation, and background removal.

Senior Software Engineer, Smart Technologies (BD) Ltd.

May 2017-Dec 2019

- Led development of enterprise supply-chain systems, including architecture and optimization of a 4 TB SQL Server deployment; reduced report generation from ~20 minutes to under one minute.

Software Engineer, Proggasoft

Mar 2015-Aug 2016

- Developed production web applications and backend systems for DevSkill.com using ASP.NET MVC and SQL.

Awards and Invited Talks

- Champion, BASIS National ICT Awards, 2020 (Retouched.ai); Finalist, Asia Pacific ICT Alliance Awards, 2021
- Artificial Intelligence in Advertising - Invited workshop speaker, Daffodil International University

Technical Skills

Methods: Conformal prediction and risk control; calibration and uncertainty evaluation; benchmark and annotation design; ablation studies and error analysis; multimodal model evaluation

Multimodal / VLM: Vision-language models, visual grounding, hallucination evaluation, multimodal reasoning, image-text alignment

Systems: Python, PyTorch, Hugging Face, FastAPI, Docker, Git, Google Cloud Platform; COLMAP, Open3D, NeRF, 3D Gaussian Splatting